SOP-TESTING, VALIDATION AND ACCEPTANCE

[CONFIDENTIAL]
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CHANGE HISTORY

This section describes the details of changes that have resulted in the current document.

SL. No.	Document Version No.	Change Description	Reason For change	Prepared By & Date	Reviewed By & Date	Approved By & Date
01	0.1	Draft	Implementation of the Integrated Management System	Sushmitha S 04-Jul-22		
02	Issue B Version 1.0	Revised the Document to suit the requirements for implementation of the integrated management system			Megha VK 06-Jul-22	Rama G 08-Jul-22
03	Issue B Version 1.1	Section 9.1 (point 7) Defect Management added	Defect Lifecycle process was missing	Anitha N 07-Nov-23	Santosh V 8-Nov-23	Santosh V 8-Nov-23
04	Issue B Version 1.2	Document formatting, font style to Poppins and alignment	Updated in line with the document guidelines	Devansh S 20-Dec-24	Anitha N 23-Dec-2024	Santosh V 31-Dec-2024
05	Issue C Version 1.0	Review of document	New version update of IMS	Anitha N 01-08-2025	Santosh V 08-Aug-25	Sanmangala MS 14-08-2025

1 Purpose

To establish a documented procedure to inspect and test the software item to complete the software project/product and to establish a standard procedure to define acceptance procedure for the software developed.

2 Scope

This applies to all the projects executed in Utthunga Technologies.

3 Abbreviations

1. PM: Project Manager
2. CI: Configurable Items
3. TSR: Test Summary Report
4. TRR: Test Result Report

4 Terms and Definitions

1. NIL.

5 Responsibilities

1. Execute full software development life cycle (SDLC)
2. Develop flowcharts, layouts and documentation to identify requirements and solutions.
3. Write well-designed, testable code.
4. Produce specifications and determine operational feasibility.
5. Integrate software components into a fully functional software system.
6. Design, develop and execute unit test plans, test designs, test cases and test strategies.
7. Design, develop and execute subsystem test plans, procedures and processes.
8. Document all test plans, test cases and strategies procedures and issues.
9. Design and implement test scripts on test tools and scripting languages.
10. Coordinate and collaborate with outside test partners.
11. Design, develop and implement program and process improvements.
12. Design and develop coding, code reviews, unit testing and release management.
13. Develop design specifications in accordance with business requirements and issues.
14. Recommend strategic improvements to optimize performances.
15. Perform analyses and interpretations of strategies and software applications.

6 RACI

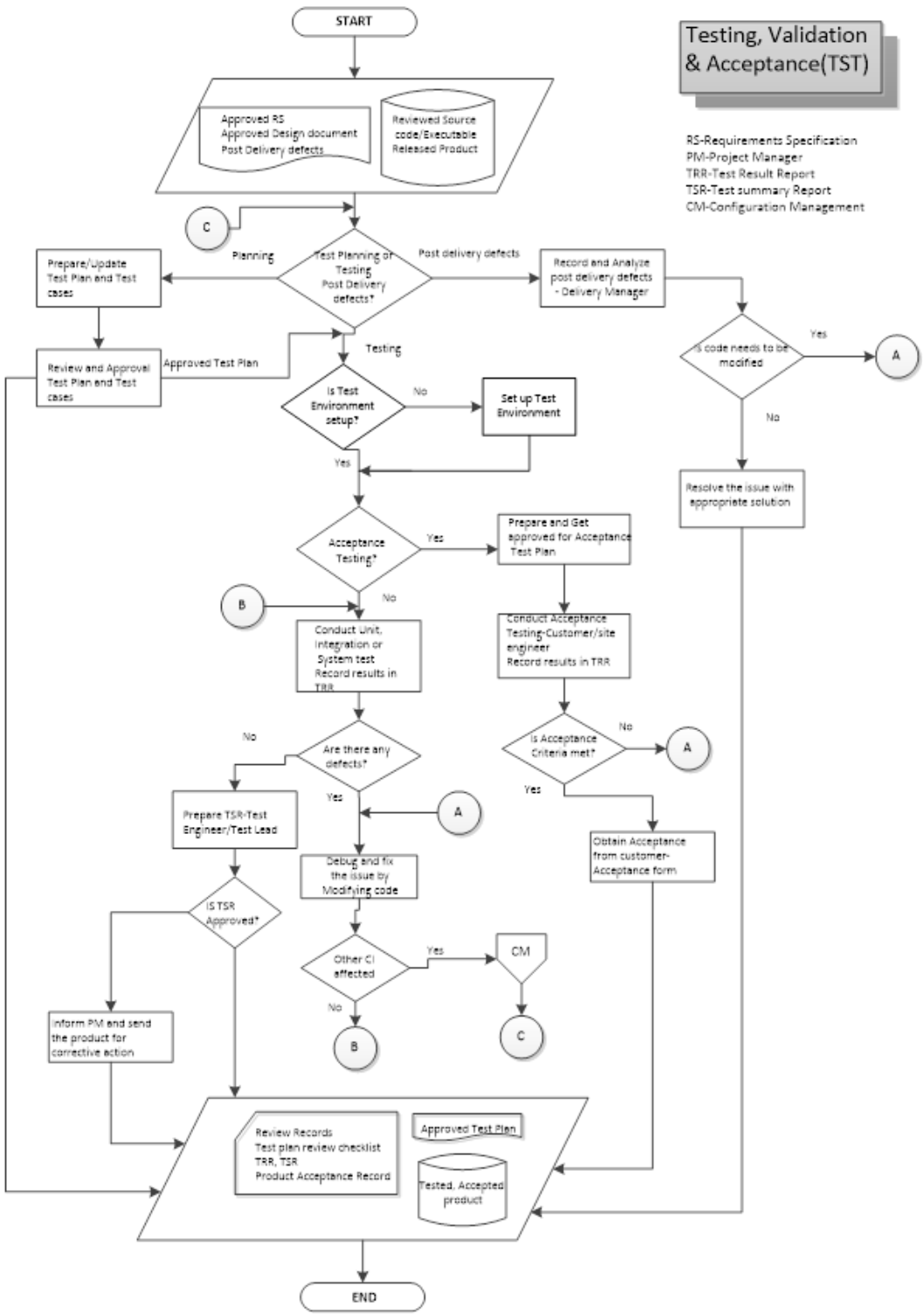
Process	Responsible	Accountable	Consulted	Informed
Testing Validation and Acceptance	Project team members	Project Manager	Head Quality	Customer / Top Management

7 SIPOC

Supplier	Input	Process and Control		Output	Customer
		Process	Control		
Customer / Internal Process / Quality	Approved RS / Design Reviewed Code and/or executable	Testing Validation and Acceptance	Checklist based review	Reviewed and Approved Test plans Approved Test Result Report, Test Summary Report, Tested Source Code / Executables Acceptance Report	Customer / Internal Process

8 Overall Process Flow Chart

a) Testing, Validation and Acceptance



9 Process Description

9.1 Testing, Validation and Acceptance

a. General

- Control of conforming and non-conforming test plan documents shall be implemented as defined in process Configuration Management and Control of Non-Conforming Product respectively.
- The development team shall make internal release to the testing team and testing shall be carried as per test plan.
- Test Environment shall be planned and shall make sure that Test environment is identified for the hardware/software. Test environment shall also define the device (CSI) identification its verification and calibration or configuration methods.
- Testing could result in the following types of defects.
  - Critical – Application crash / show-stopper defects i.e. there is no workaround solution available.
  - Major – Non-conformance to requirements / specifications
  - Minor - Cosmetic Errors
- The records related to the test plan, test result report, and test summary report shall be maintained by Project Manager or person designated by Project Manager.

b. Test Planning

- The Project Manager or person designated by Project Manager shall prepare the Test Plan (System / Integration / Unit Test Plans) after the Requirements / Design documents are approved.
- The Project Manager or person designated by Project Manager shall ensure that the test plan, test cases and / or the test scripts are designed as per the requirements.
- Test Plan shall be prepared as per the Test Plan Template.
- Mandatory Test cases shall also be identified.
- Test Plan Review Checklist shall be used in the review of the Test plan.
- Review findings shall be recorded in the Review Form and tracked to closure as described in the section Review Process.
- Approval shall be as defined in the section Review Process.

c. Test Environment

- Testing environment is a setup of software and hardware on which the testing team is going to perform the testing of the newly built software product.
- This setup consists of the physical setup which includes hardware, software, Operating system, or any other components required to test the software.
- Detailed test environment shall be mentioned in the Test Plan.
- Test environment shall also mention the measuring equipment/devices and the methods used to calibration/configuration prior to use to confirm the ability of them to satisfy the intended application.

d. Test Execution

- The Developer/Test Engineer shall test the product as per the test plan, test cases and / or test scripts (automation).
- Mandatory test cases shall be executed by the test engineer for every build to ensure mandatory requirements are not missed.
- The Developer/Test Engineer shall record the results of the test in the Test Result Report.
- Defects found during the testing shall be logged with all relevant details into defect tracking system. Defects are classified as defined in the section Defect Classification.
- In case of acceptance, the software item shall be certified for the next phase.
- In case the software item is rejected, the software item shall be sent for defect fixing as described in section Defect / Problem Fixing

e. Defect Classification

- All defects found during testing shall be recorded in defect tracking system.
- Defects shall be classified by the person who has reported the defect in consultation with project team. The defects shall be classified as follows:
- Critical: Critical defects are the defects, which results in the failure of the complete software system, or of a subsystem, or of a software unit so that no work or testing can be carried out after the occurrence of the defect, such as
    - Functionality does not permit for further testing.
    - Runtime Errors like JavaScript errors etc.

- Functionality Missed out / Incorrect Implementation (Major Deviation from Requirements).
- Performance Issues (If specified by Client).
- Browser incompatibility and operating systems incompatibility issues depending on the impact of error.

- Major: A defect, which will cause an observable product failure or deviating from requirements, such as
  - Functionality incorrectly implemented (Minor Deviation from Requirements).
  - Performance Issues (If not specified by Client).
  - Mandatory Validations for Mandatory Fields.
  - Images, Graphics missing which hinders functionality.
  - Front End / Home Page Alignment issues.
- Minor: Minor defects are small errors that do not prevent or hinder functionality.
  - Screen Layout Issues
  - Spelling Mistakes / Grammatical Mistakes.
  - Documentation Errors
  - Page Titles Missing.
  - Alt Text for Images.
  - Background Color for the Pages other than Home page.
  - Default Value missing for the fields required.
  - Cursor Set Focus and Tab Flow on the Page.
  - Images, Graphics missing, which does not, hinders functionality.

f. Defect / Issue Fixing

- Defects identified during testing by the testing team shall be communicated to the project team by way of Test Result Report.
- If the defects result in changes to other configurable items (CI), then the changes shall be handled as defined in Configuration Management process.
- Code shall be modified for defect fixing.
- Binaries/Executables shall be released for testing team for test execution as described in section Test Execution for the conformity of the product.

g. Defect Lifecycle

- The defect is logged into a defect tracking system or issue management tool. This report includes details such as the defect's description, steps to reproduce it, and any related files or screenshots.
- The defect is assigned to a developer or a development team responsible for fixing it.
- The defect report is communicated to the developer with all the necessary details.
- The defect's status is updated in the defect tracking system. Common statuses include:
  - New: The defect has been identified but not yet assigned.
  - Assigned: The defect has been assigned to a developer.
  - In Progress: The developer is actively working on fixing the defect.
  - Fixed/Resolved: The developer has fixed the defect, and it's ready for verification.
  - Reopened: The defect is reopened if the fix did not resolve the issue or introduced new problems.
  - Closed: The defect is considered resolved and closed.

h. Test Summary Report

- The Test Result Report shall be consolidated and recorded in the Test Summary Report. Test Summary Report shall be documented for Unit, Integration, System and Acceptance tests.
- Test Engineer / Test Leader shall prepare the TSR.
- The Test Manager and Project Manager shall decide to accept / reject the product for release to the Customer based on the Test Summary Report (TSR). Test Manager shall approve the

i. Acceptance Planning

- The Project Manager or person designated by Project Manager shall assist the Customer to define the acceptance test plans.
- Acceptance test plan, wherever applicable, shall be prepared as per the Test Plan Template.
- Review of Acceptance test plan shall follow as defined in the procedure Review Process
- The acceptance criteria may be agreed upon during the contractual agreement. The following may be decided during the contract:
  - Schedule

- Resource requirement
- Location
- Evaluation procedure
- Acceptance Criteria

- However, any change shall be subsequently discussed and agreed upon.

j. Acceptance Testing by Customer

- The Customer shall carry out acceptance testing. Any assistance required by the Customer during acceptance testing shall be provided.
- The results of the tests shall be recorded in the Test Result Report.
- Acceptance can be received from Customer in a form of written communication or through Acceptance Form.

k. Acceptance Testing by Site Engineer

- If required Site Engineer from Utthunga shall carry out the Acceptance Testing.
- The results of the tests shall be recorded in the Test Result Report.
- Product Acceptance shall be received from the Customer through the Acceptance Form.

l. Failures during Acceptance Testing

- The Project Manager or person designated by Project Manager shall decide the corrective action against the failures observed during the acceptance testing within the time frame as decided and documented in consultation with the Customer.
- All identified defects shall be traced to the code.
- On completion of the corrective action by the Project Manager or person designated by Project Manager, acceptance testing shall be carried out again.

m. Defect Analysis

- Following are the list of criteria for conducting the analysis.

Example Activities	Further Explanation
Compare actual results to expected results	
Identify the results that do not meet established criteria for verification	

Identify the results that do not meet established criteria for validation	
Analyze verification and validation results that do not meet established criteria and determine corrective action	Results may include defect, error or issue analysis
Submit improvement proposals where improvement to verification and validation process are identified	
Record and communicate defect analysis results and corrective actions to affected stakeholders	

n. Post Delivery defects/defects tracking and fixing.

- Post-delivery defects are received by Delivery Manager/Project Manager from customer.
- Received defects shall be recorded and analyzed by the Project Manager
- If the defect result in changes to other configurable items (CI), then the changes shall be handled as defined in Configuration Management process.
- Defects reported by Customer shall be handled as defined in Control of Non-Conforming process.
- All Customer identified defects shall be traced to the code.

10 Information Security requirements

a. Procedures shall be implemented for the management of removable media in accordance with the classification scheme adopted by the organization.

b. Media containing information shall be protected against unauthorized access, misuse, or corruption during transportation.

c. Development, testing, and operational environments shall be separated to reduce the risks of unauthorized access or changes to the operational environment.

d. The information security related requirements shall be included in the requirements for new information systems or enhancements to existing information systems.

e. Information involved in application services passing over public networks shall be protected from fraudulent activity, contract dispute and unauthorized disclosure and modification.

f. Information involved in application service transactions shall be protected to prevent incomplete transmission, misrouting, unauthorized message alteration, unauthorized disclosure, unauthorized message duplication or replay.

g. Rules for the development of software and systems shall be established and applied to developments within the organization.

h. Changes to systems within the development lifecycle shall be controlled using formal change control procedures.

i. When operating platforms are changed, business critical applications shall be reviewed and tested to ensure there is no adverse impact on organizational operations or security.

j. Modifications to software packages shall be discouraged, limited to necessary changes and all changes shall be strictly controlled.

k. Principles for engineering secure systems shall be established, documented, maintained, and applied to any information system implementation efforts.

l. Organizations shall establish and appropriately protect secure development environments for system development and integration efforts that cover the entire system development lifecycle.

m. The organization shall supervise and monitor the activity of outsourced system development.

n. Testing of security functionality shall be carried out during development.

o. Acceptance testing programs and related criteria shall be established for new information systems, upgrades, and new versions.

p. Test data shall be selected carefully, protected, and controlled.

q. Confidentiality of data is maintained.

11 References

a. ISO 9001 Clauses 8.5,8.6,8.7

b. ISO 27001 Clauses 8.1, Controls A.6, A.7, A.9, A.10, A.12, A.14

c. ISMS Control Policies

d. Acceptance Form

12 Annexures

NIL

13 Forms

a. SDL Security Verification Checklist UT/SD/TVA/TMP/01

b. SDL Test Plan Review Checklist UT/SD/TVA/TMP/02

c. SDL Test Plan Template UT/SD/TVA/TMP/03

d. Test Case Review Checklist UT/SD/TVA/TMP/04

e. Test Plan Review Checklist UT/SD/TVA/TMP/05

f. Test Plan Template UT/SD/TVA/TMP/06

g. Test Result Report UT/SD/TVA/TMP/07

h. Test Summary Report UT/SD/TVA/TMP/08

i. Unit Test Report UT/SD/TVA/TMP/09